

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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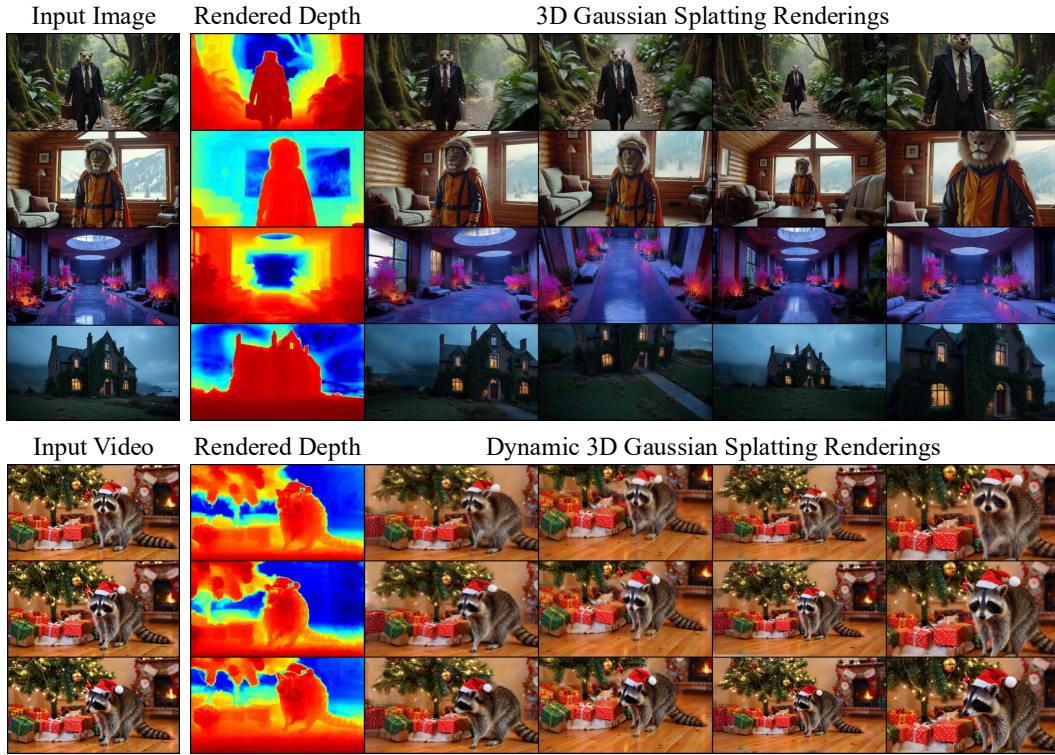


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** Given a single image (top), *Lyra* predicts a 3D Gaussian Splatting (3DGS) representation in one feed-forward pass by self-distilling a video diffusion model, without using real-world multi-view data. Given a video input (bottom), *Lyra* predicts a dynamic 3DGS representation that supports interactive control over both time (rows) and viewpoint (columns).

ABSTRACT

Generating virtual environments is a central capability for applications that span interactive entertainment and physical-AI domains, including robotics, autonomous driving, and industrial AI. Existing learning-based 3D reconstruction methods typically require captured real-world multi-view data, and such data are often unavailable, costly to acquire, or insufficiently diverse. Recent video diffusion models exhibit strong generative capacity over realistic image sequences; however, their outputs remain two-dimensional, which restricts their direct use in simulation settings where an embodied agent must navigate, render, and interact with a spatially explicit environment. In this paper, we introduce a self-distillation framework that transfers the implicit 3D information encoded in video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, thereby removing the requirement for multi-view training captures. Specifically, we augment the standard RGB decoder with a 3DGS decoder and use the RGB decoder output as supervision for the 3D branch. Under this formulation, the 3DGS decoder can be trained entirely from synthetic supervision generated by video diffusion models.

At inference time, the resulting model synthesizes 3D scenes from either a text prompt or a single image and supports real-time rendering. The same framework further extends to dynamic 3D scene generation from a monocular input video. Experiments demonstrate that the proposed framework achieves state-of-the-art performance for both static and dynamic 3D scene generation.

1 INTRODUCTION

Scalable construction of high-quality 3D environments remains a fundamental problem in computer vision and graphics, with direct relevance to film production, VR/AR, and closed-loop physical-AI simulation. These settings require explicit 3D scene representations that support real-time rendering, physically meaningful interaction, and view-consistent synthesis across camera trajectories.

Recent neural 3D reconstruction methods (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) can recover such representations from posed images. Their applicability, however, is constrained by the requirement for accurate camera poses and high-quality image observations. The constraint becomes more severe for dynamic scenes, where synchronized multi-camera capture is typically required (Li et al., 2022). Reconstruction methods are additionally limited to the observed scene content and therefore cannot reliably infer unobserved regions outside the input views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) seek to reduce these limitations, but they introduce a separate bottleneck: large and diverse 3D training corpora are scarce, which degrades generalization outside the training distribution.

Video diffusion models (Cosmos, 2025; Wan, 2025) provide a complementary source of supervision. Because they are trained on internet-scale video corpora, these models exhibit high visual fidelity and broad generalization across environments. Their training data contain diverse camera motions, including handheld trajectories, cinematic pans, and aerial footage, and therefore induce implicit information about scene geometry and visibility without requiring explicit multi-view 3D annotation. Unlike reconstruction-only approaches (Charatan et al., 2024), generative video models can synthesize plausible content beyond the observed input frames. Nevertheless, their native output is a sequence of 2D images rather than an explicit 3D representation, limiting direct use in simulation and other downstream applications that require geometric consistency, long-horizon coherence, and interaction with spatial structure. Recent work (Wang et al., 2024e; Ren et al., 2025) shows that video models can be adapted for explicit camera control using relatively small posed datasets. Camera-controllable video synthesis therefore provides both structured inputs and supervisory signals for training 3D reconstruction models.

Motivated by this observation, we connect reconstruction and generation through the task of *generative 3D scene reconstruction*. We propose *Lyra*, a method that produces explicit 3D environments from the latent representations of a video diffusion model in a single forward pass. The central mechanism is a self-distillation framework (Fig. 2) in which a 3D Gaussian Splatting (3DGS) decoder operates directly in the video diffusion latent space. Given a single input image or video, we sample a camera trajectory as conditioning information, denoise the corresponding video latent, and decode the latent through two parallel branches. The standard RGB decoder synthesizes a video sequence, while the proposed 3DGS decoder predicts an explicit 3D representation. These branches define a teacher-student training objective: the RGB branch provides synthetic image supervision, and the 3DGS branch learns to produce a renderable scene representation. To increase viewpoint coverage, we sample multiple camera trajectories, generate multiple video latents, and train the 3DGS decoder to fuse the resulting information while reducing long-range and multi-view inconsistencies.

This formulation provides three technical advantages: **(i)** it enables large-scale synthetic environment generation across diverse scenarios directly from video diffusion models, avoiding dependence on real-world multi-view capture; **(ii)** by operating in the latent space of a video model, it permits efficient processing of multiple views without the memory cost associated with pixel-space feed-forward reconstruction; and **(iii)** by outputting 3DGS primitives, it yields an explicit representation with geometric consistency and immediate compatibility with downstream tasks such as physical simulation. Consequently, at inference time, *Lyra* generates high-quality 3D Gaussian scenes from monocular input that support real-time rendering, *without requiring any additional optimization or post-processing*.

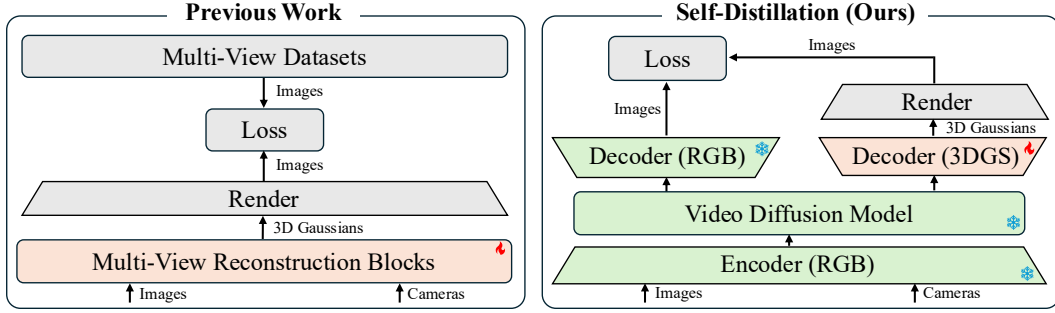


Figure 2: **Self-distillation framework of *Lyra***. Prior work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks on real-world datasets with limited diversity (Zhou et al., 2018; Ling et al., 2024b). Our self-distillation framework (right) instead uses a pre-trained camera-controlled video diffusion model as the teacher for generative 3D scene reconstruction. The RGB decoder output supervises renderings from the 3DGS decoder, and the diversity of the video pre-training data provides broad synthetic multi-view supervision.

We further extend the same self-distillation principle to dynamic 3D scenes. In this setting, the input is a monocular video rather than a single image, and the model predicts time-dependent 3D Gaussian representations. The extension preserves the core teacher–student structure while conditioning the decoder on source and target time embeddings. We also introduce a dynamic data augmentation procedure that balances supervision across timestamps and viewpoints, preventing the decoder from collapsing to timestep-local information when reconstructing motion.

Our experimental evaluation verifies the proposed design on static and dynamic generation benchmarks. Across RealEstate10K, DL3DV, Tanks-and-Temples, and our generated *Lyra* datasets, the method improves quantitative metrics and visual quality relative to prior single-image-to-3D and monocular-video-to-4D baselines. Ablations further isolate the effects of self-distillation, latent-space decoding, multi-view fusion, depth supervision, opacity pruning, and the dynamic augmentation strategy.

In summary, our contributions are:

- We formulate generative 3D scene reconstruction as self-distillation from camera-controlled video diffusion models into an explicit 3DGS decoder.
- We introduce a latent-space 3DGS decoder that reconstructs 3D Gaussian scenes from video diffusion latents in a single forward pass, without per-scene optimization or post-processing.
- We extend the framework to dynamic 3D scene generation from monocular video using time-conditioned decoding and dynamic data augmentation.
- We demonstrate state-of-the-art results on static and dynamic 3D generation benchmarks and provide ablations that quantify the contribution of the principal architectural and training components.

2 RELATED WORK

Multi-view image generation. Early multi-view image generation methods (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) primarily address object-centric settings and typically omit scene backgrounds. Subsequent approaches (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) extend the generation domain from isolated objects to scene-scale content. CAT3D (Gao et al., 2024b) adapts a pre-trained image diffusion model to generate multi-view images from either generated or real image inputs. These generated views are then reconstructed into an explicit 3D representation, such as NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023), in a separate optimization stage. Related methods further extend this paradigm to dynamic 3D scenes by generating multiple viewpoints with space-time diffusion models (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Although these pipelines produce strong visual results, converting their generations into an explicit 3D representation requires costly per-scene optimization that is not amortized across test instances. Our objective differs from this generate-then-optimize design: we target *direct* generation of a 3D scene from text or from a single image.

Camera-conditioned video generation. A substantial line of work fine-tunes video diffusion models to support 3D camera control. MotionCtrl (Wang et al., 2024e) introduced this direction by condi-

tioning pre-trained video models on camera poses. Follow-up methods (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) encode cameras as Plücker coordinates for pixel-wise conditioning, while training-free alternatives (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) modify video diffusion models without additional camera-control training. More recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) demonstrate high-quality camera-controlled video synthesis using large-scale synthetic training data. These methods provide controllable 2D video outputs, but the generated sequences are not themselves grounded in an explicit 3D representation. Our work is orthogonal to this line: we use camera-controlled video models as supervision for feed-forward 3D generation, so that novel-view rendering is available *after* generation. Concretely, we build on GEN3C (Ren et al., 2025), a camera-controlled video diffusion model, and instantiate it as the teacher in a self-distillation framework for 3D reconstruction.

Feed-forward 3D models. Early feed-forward 3D reconstruction methods were also largely object-centric and predicted either a NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) representation from text or from an image. More recent scene-level reconstruction models (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) typically regress per-pixel 3D Gaussians from one or more input images. However, most scene-level models remain closely tied to the distribution of their training data, such as RealEstate10K (Zhou et al., 2018), and therefore generalize poorly to generated or out-of-distribution scenes. Recent methods adapt feed-forward 3D reconstruction toward scene generation. Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder to generate pointmaps together with multi-view images, which are then used for feed-forward 3D reconstruction. Most closely related to our setting, Wonderland (Liang et al., 2025a) uses a camera-controlled video model to generate 3D Gaussians with a feed-forward network. In contrast, we use the camera-controlled video model as a teacher in a self-distillation procedure and train the student 3D model *without requiring real-world multi-view data*. We additionally show that the same formulation admits a simple feed-forward extension to 4D scene generation, a task not previously explored in this setting.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

The proposed method transfers the implicit 3D structure represented by a video diffusion model into an explicit 3DGS decoder that produces renderable scene geometry. We instantiate this transfer as a teacher–student system: the camera-controlled video diffusion model acts as the teacher by generating RGB videos, and a student 3DGS decoder, operating in the same latent space, is optimized to reproduce those teacher outputs as explicit Gaussian primitives, as shown in Fig. 2. This section first specifies the video diffusion backbone (Sec. 3.1) and then describes its use in the self-distillation objective for the 3DGS decoder (Sec. 3.2). We subsequently define the 3DGS decoder architecture in Sec. 4 and describe the minimal modifications required for dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because the 3DGS decoder is supervised exclusively by a camera-controlled video diffusion model, the geometric consistency of that teacher is a critical assumption. We therefore base the framework on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model with explicit mechanisms for 3D-aware conditioning. We summarize the components of GEN3C that are used by our method.

Spatiotemporal 3D cache. To improve temporal coherence and camera-control accuracy, GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from the input image(s) or videos. Each cache element $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimate (Wang et al., 2024a) of an RGB image observed from camera viewpoint v at time t . The cache has an $L \times V$ organization, where L is the temporal sequence length and V is the number of camera views.

Rendering and structured guidance. GEN3C uses the cache by rendering each point cloud from specified camera poses, yielding RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ through $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$. Here, $\mathcal{R}$ denotes the rendering operator that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane according to camera pose $\mathbf{C}^t$ at time t . For a camera sequence $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache entries produces V videos of length L . These renderings are stacked into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks identify pixels for which the diffusion model must synthesize

previously unseen content, and the rendered images provide structured visual conditioning for the subsequent video-generation step.

Video variational autoencoder (VAE). Diffusion-based video generators typically use a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to map high-dimensional videos into a compressed latent representation for efficient training and sampling. Given an RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, the pre-trained VAE encoder $\mathcal{E}$ maps the video to $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$. Both diffusion training and inference are performed in this latent space, and the final RGB sequence $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is recovered by the pre-trained VAE decoder $\mathcal{D}_{rgb}$. We use the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model. Its latent channel dimension is $C = 16$, its compressed temporal dimension is $L' = (L - 1)/\tau + 1$, and its spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, with temporal compression factor $\tau = 8$ and spatial compression factor $\sigma = 8$.

3.2 SELF-DISTILLATION

We optimize the 3DGS decoder $\mathcal{D}_s$ as the student in a teacher–student training procedure, where the camera-controlled video diffusion model $\mathcal{V}$ provides all image supervision. To construct diverse teacher inputs, we curate a large-scale prompt set with large language models (OpenAI, 2025; Bai et al., 2023), generate images I with an image diffusion model (Black Forest Labs, 2024), and expand each image into multi-view video sequences using GEN3C (Ren et al., 2025).

Teacher–student setup. Given an input image I and sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ produces a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. The latent $\mathbf{z}$ is decoded through two branches. The pre-trained RGB decoder $\mathcal{D}_{rgb}$ generates teacher frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ predicts explicit 3D Gaussians $\mathbf{G}$. Student renderings are defined as $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and are optimized to match the teacher frames $\mathbf{I}_{\mathcal{D}_{rgb}}$, which closes the self-distillation loop.

Multi-trajectory supervision. To increase viewpoint coverage around each input image, we sample $V = 6$ camera trajectories per image, as illustrated in Fig. 3. For trajectory v , we build a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ and associate it with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through $\mathcal{V}$ yields trajectory-specific latents $\mathbf{z}^v$, and the teacher branch decodes them into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The decoder $\mathcal{D}_s$ is trained to fuse the multiple $\mathbf{z}^v$ tensors into a single coherent Gaussian representation $\mathbf{G}$ while completing disoccluded regions. This procedure trains $\mathcal{D}_s$ entirely from synthetic supervision supplied by $\mathcal{V}$. Latent-space aggregation makes the multi-trajectory input computationally tractable, whereas the explicit Gaussian output $\mathbf{G}$ enforces a renderable 3D representation suitable for downstream simulation and real-time rendering.

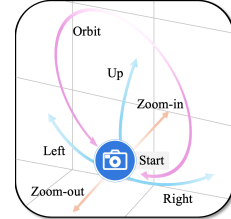


Figure 3: Six sampled camera trajectories used to maximize viewpoint coverage.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

The feed-forward static decoder $\mathcal{D}_s$ maps synthesized multi-view latents from the video model $\mathcal{V}$ to an explicit 3D representation that can be rendered from arbitrary viewpoints. We use 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) because it provides an explicit primitive-based representation with efficient rasterization. Fig. 4 (right) visualizes the complete 3DGS decoder.

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Previous feed-forward reconstruction systems predict 3D Gaussians at the pixel level and are therefore limited by the number and resolution of input images that can be processed. For example, GS-LRM (Zhang et al., 2024e) uses 2–4 images at 512×512 resolution, whereas AnySplat (Jiang et al., 2025a) is trained with 24 images at 448×448 resolution. Our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution, which exceeds the memory capacity of existing pixel-space reconstruction architectures. The dominant bottleneck is attention over visual tokens, whose memory and compute scale with pixel count. We avoid this scaling regime by performing reconstruction directly in the compressed video latent space

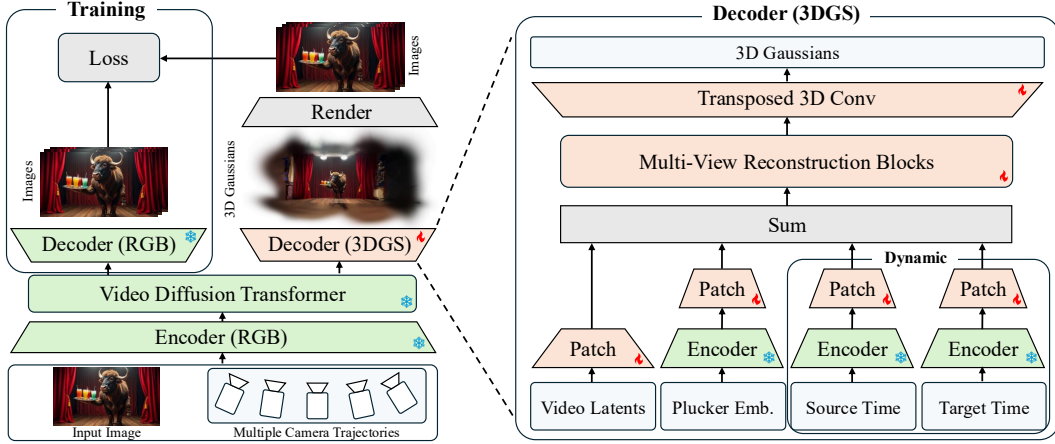


Figure 4: **3D Generative Reconstruction Framework.** The pipeline uses a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large-scale data. We train a 3D Gaussian Splatting (3DGS) decoder by matching renderings of predicted 3DGS scenes to the RGB-decoded outputs of the frozen video model. Only the 3DGS decoder is optimized; the pre-trained autoencoder and diffusion model remain fixed. At inference time, the RGB decoder is no longer required and the 3DGS decoder directly produces the scene representation. Time conditioning in the 3DGS decoder extends the same architecture from static to dynamic 3D scene generation.

produced by the camera-controlled video diffusion model. The resulting multi-view video latents are denoted $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ maps its latent inputs to the hidden dimension of the reconstruction network and predicts per-pixel Gaussian parameters $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Its inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, with $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input tensor is patchified to match the reconstruction-block hidden dimension. No additional visual encoder is required; a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) is sufficient to convert the video latents into flattened reconstruction tokens. The network processes the sum of the latent and Plücker tokens with reconstruction blocks following Long-LRM (Ziwen et al., 2024): each block contains one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers. We repeat this block twice, yielding 16 layers with hidden dimension 512. A final transposed 3D convolution maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. Raw Plücker embeddings are computed from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. We reuse the RGB encoder $\mathcal{E}$ from the pre-trained VAE to encode these Plücker embeddings. Specifically, the 3-dimensional ray directions and the 3-dimensional cross products between ray directions and origins are encoded separately with $\mathcal{E}$ and then concatenated along the channel dimension, producing $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of two so that it matches the video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. The 3DGS decoder is supervised with image-space reconstruction losses computed between rendered student views and teacher RGB frames. The objective consists of a Mean Squared Error (MSE) term $\mathcal{L}_{mse}$ and an LPIPS term $\mathcal{L}_{lips}$ (Zhang et al., 2018) using VGG (Simonyan & Zisserman, 2014) as the feature extractor.

Depth supervision. Training with RGB supervision alone often yields flattened geometry. We therefore add depth supervision by comparing rendered depth maps to temporally consistent video depth estimated by ViPE (Huang et al., 2025), using the scale-invariant depth loss $\mathcal{L}_{depth}$ from Long-LRM (Ziwen et al., 2024).

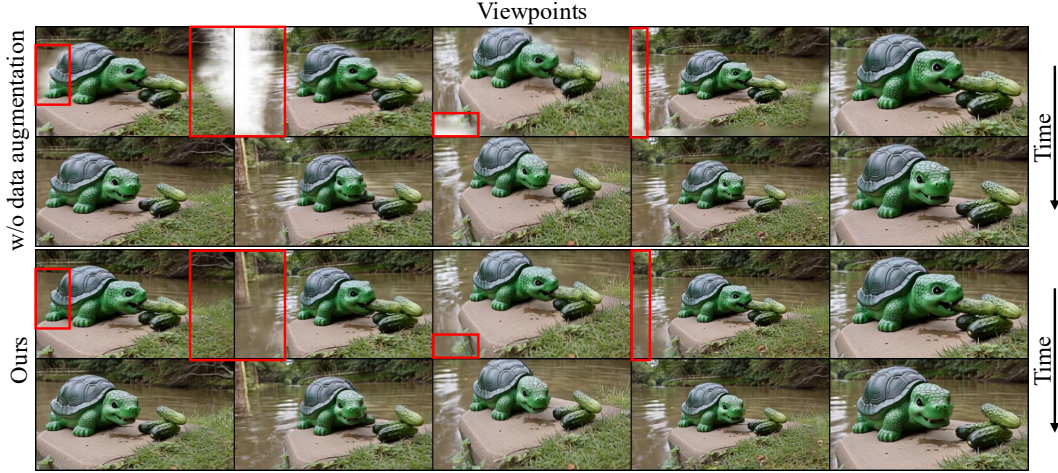


Figure 5: **Dynamic data augmentation.** Naive supervision of the time-aware 3DGS decoder produces artifacts in the generated Gaussians: early timesteps observed from extreme poses assign low opacity to regions outside the available supervision. We augment supervision with a motion-reversed video so that every timestep receives broader spatial coverage, which suppresses low-opacity artifacts at early timesteps.

Opacity-based pruning. Following Long-LRM (Ziwen et al., 2024), we apply L1 regularization to opacity through $\mathcal{L}_{opacity}$ and prune the 80% of Gaussians with the lowest opacity values.

Total loss. The total objective is the weighted sum of the reconstruction, perceptual, depth, and opacity terms, with scalar coefficients λ_i :

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

The same framework extends to dynamic 3D scenes by adding time conditioning while preserving the static pipeline structure.

Self-distillation for dynamic scenes. The dynamic setting follows the static teacher–student construction with a video input in place of a single image. The video model receives a monocular video of length L together with corresponding camera poses and generates multi-view video latents that represent the same underlying motion. For video inputs, we use the same data-generation protocol as in the static case: diverse text prompts are sampled with large language models, videos are generated with video diffusion models (Cosmos, 2025; Wan, 2025), and the generated videos are annotated with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. We adopt the bullet-time formulation of Liang et al. (2025b) and define a dynamic 3DGS decoder $\mathcal{D}_s$ that predicts time-dependent 3D Gaussians for all video frames. Architecturally, the dynamic 3DGS decoder $\mathcal{D}_s$ follows $\mathcal{D}_s$ except that the $\mathcal{D}_s$ input is augmented with encoded source and target time embeddings $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The raw source time tensor assigns a time value to each input frame, yielding $\mathbf{T}^{src, raw} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the raw target time tensor is $\mathbf{T}^{tgt, raw} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both tensors are spatially repeated, augmented with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, and encoded by the RGB encoder $\mathcal{E}$ into latent tensors $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time tensor is repeated along the latent temporal dimension to match L' . During training, we randomly sample a target timestep, decode the corresponding 3D Gaussians, and supervise the rendered outputs across all trajectories produced by dynamic data augmentation. We initialize $\mathcal{D}_d$ from the pre-trained $\mathcal{D}_s$, with the patchification layers for $\mathbf{T}^{src}$ and $\mathbf{T}^{tgt}$ initialized to zero. The tensors $\mathbf{T}^{src}$ and $\mathbf{T}^{tgt}$ are added directly to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{src}, \mathbf{T}^{tgt})$.

Dynamic data augmentation. For static 3D scenes, all frames across timesteps can supervise the same 3DGS representation because the scene geometry is time-invariant. For dynamic scenes, the

Table 1: **State-of-the-art comparisons.** The table compares our method with prior single-image-to-3D generation methods on RealEstate10K, DL3DV, and Tanks-and-Temples.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336

3DGS representation is time-dependent, so only frames at the matching timestamp provide valid supervision. This restriction can induce a degenerate solution in which the 3DGS at a given timestep ignores information from other timesteps (see Fig. 5).

We address this imbalance with a dynamic data augmentation strategy that distributes supervision more uniformly over timesteps. Early frames naturally correspond to viewpoints close to the input image, whereas later frames correspond to more distant viewpoints. To provide both near and far supervision at each timestep, we augment the training data with paired supervision views. Concretely, we reverse the input video temporally and pass it through the video model $\mathcal{V}$, which generates six additional multi-view sequences. After reversing these sequences back to the original temporal order, we obtain six inward trajectories that move from distant viewpoints toward the input. Combining these inward trajectories with the original six outward trajectories yields 12 supervision views per timestep. This augmentation is used only during training: the reversed trajectories provide additional supervision that reduces collapse caused by pruning artifacts and are not required during inference.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. Training does not use existing real-world multi-view datasets. Instead, we construct the *Lyra* dataset and use the video model as the sole source of supervision for the 3DGS decoder. The static 3D setting contains 59,031 generated images, and the dynamic 4D setting contains 7,378 generated videos. All samples are derived from diverse text prompts that cover indoor and outdoor scenes, humans, animals, realistic content, and imaginative content. For each image in the 3D setting and each video in the 4D setting, we synthesize six camera trajectories, resulting in 354,186 videos for 3D training and 44,268 videos for 4D training.

Baselines. We compare against state-of-the-art generative single-image-to-3D reconstruction methods: ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Because *no source code is available* for evaluating these methods on our out-of-distribution evaluation set, the main quantitative comparisons use the metrics reported in the corresponding papers.

Evaluation. We evaluate single-image-to-3D generation on RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017). The evaluation protocol follows Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b). Reconstruction quality is measured with the standard image metrics PSNR, SSIM, and LPIPS.

6.2 MAIN RESULTS

Quantitative results. Tab. 1 reports quantitative results for single-image-to-3D generation. The proposed method obtains the best score on every benchmark and for every reported metric. Because the 3DGS decoder is supervised by the video generator, further improvements in camera-controlled video generation are expected to transfer directly to reconstruction quality. Additional quantitative comparisons on the *Lyra* dataset are provided in Appendix C.

Qualitative results. Fig. 1 and Fig. 6 show novel-view renderings from generated 3DGS scenes; video results will be released upon acceptance. The method synthesizes plausible unseen regions for both 3D and 4D scenes while maintaining consistency with the input image or input video. Qualitative comparisons on the *Lyra* dataset are included in Appendix C.



Figure 6: **Image-to-3DGS Generation.** The figure visualizes five rendered viewpoints from generated 3DGS scenes.



Figure 7: **Ablations.** We evaluate each ablation by rendering the generated 3DGS scene from the same extreme novel viewpoint; depth visualizations and the corresponding depth ablation are reported in Appendix C.4.

6.3 ABLATIONS

Tab. 2 evaluates the contribution of the principal design components on out-of-distribution diverse prompts. Each ablation compares rendered 3DGS outputs against the videos generated by the camera-controlled video diffusion model.

Real data only. This variant removes self-distillation and trains only on real multi-view datasets, namely RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b). Although these datasets are standard in prior reconstruction work, they provide insufficient coverage for the diverse out-of-distribution scenes generated by our pipeline.

Self-distillation + real data. Joint training on self-distilled supervision and real data does not improve over training with self-distillation alone. This result indicates that the synthetic teacher supervision is sufficiently diverse and consistent to train the reconstruction model without additional real multi-view data.

No depth loss. Removing the depth loss increases the tendency toward flattened geometry and slightly decreases image-based metrics. Depth-map visualizations are provided in Appendix C.4.

No opacity pruning. This variant retains all Gaussians instead of pruning them according to opacity. Opacity pruning produces a more compact representation and slightly improves visual quality. At $H = 704$ and $W = 1280$, rendering takes 18ms with pruning and 30ms without pruning, corresponding to a $1.67\times$ speedup.

Table 2: **Ablation study on the *Lyra* dataset.**

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS			OOM

No LPIPS loss. The LPIPS term improves robustness to input inconsistencies and better preserves high-frequency image detail.

No multi-view fusion. A simple alternative is to predict 3D Gaussians independently for each camera trajectory and merge the resulting Gaussians into one point cloud. Learned multi-view fusion in the reconstruction blocks performs substantially better because tokens from different trajectories can exchange information through attention.

No Mamba-2. Replacing the joint Transformer/Mamba-2 design with Transformer-only blocks yields lower quality and substantially higher latency. For $V = 6$, $L = 121$, $H = 704$, and $W = 1280$, one forward pass requires 3213ms with the joint blocks and 20922ms with Transformer-only blocks, giving a $6.5 \times$ speedup for the proposed architecture.

No latent-based 3DGS. Pixel-space 3DGS decoders, such as BTimer (Liang et al., 2025b), are constrained to small numbers of input frames; BTimer is restricted to 12 input frames, whereas our latent-space decoder can use up to 726 input frames. Replacing latent-space decoding with pixel-space decoding therefore exceeds memory limits.

7 CONCLUSION

We presented *Lyra*, a unified framework for static 3D and dynamic 4D generation from a single image or monocular video. Rather than relying on collected multi-view datasets, the method uses a camera-controlled video diffusion model as a teacher in a self-distillation procedure for a student 3DGS decoder. The decoder operates in the video-model latent space and predicts explicit 3D Gaussians directly, without per-scene optimization or post-processing. This design improves scene coverage and out-of-distribution generalization, and the dynamic extension demonstrates that the same mechanism can generate time-conditioned 3DGS representations from monocular video. The scale and consistency of the generated scenes are currently bounded by the capacity of the underlying camera-controlled video diffusion model. Accordingly, improvements in video diffusion modeling should translate into larger and more consistent scene synthesis within our framework. Incorporating auto-regressive techniques (Chen et al., 2024a) is a promising direction for extending generation to larger spatial extents. Finally, integrating explicit motion and tracking information into the reconstruction network, as explored in concurrent work (Lin et al., 2025), may further improve dynamic visual quality.

8 ETHICS STATEMENT

As with other generative AI systems, the proposed method could be misused to create misleading or synthetic 3D/4D content. The research objective of this work is to support simulation, robotics, and embodied AI by providing scalable and controllable mechanisms for generating training and evaluation environments. We therefore expect the primary technical contributions to arise from the following capabilities.

- **The ability to enable realistic simulation for embodied agents:** The method provides controllable and physically consistent camera motion in generated 3D and 4D environments, supporting the training and evaluation of agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** The framework removes the requirement for multi-view capture and expensive per-scene optimization, enabling high-fidelity and customizable scene generation at scale for robotics, reinforcement learning, and closed-loop simulation research.
- **Better technical understanding:** As an academic study, the work provides a controlled case study of how camera control interacts with spatiotemporal generative models, with implications for research in computer vision, graphics, and embodied AI.

We caution against deceptive or harmful use cases in which realistic generative outputs are used to produce misleading or unverifiable media. Safeguards such as provenance tracking, dataset documentation, and rigorous evaluation protocols can help ensure that generative 3D and 4D models are developed and deployed in ways that benefit scientific and societal applications.

9 REPRODUCIBILITY STATEMENT

To support full reproduction and facilitate subsequent research, we will release the training and inference code, model weights, and data for both 3D and 4D generation upon acceptance.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section gives additional details about the GEN3C (Ren et al., 2025) video diffusion backbone introduced in Sec. 3.1 and describes the refinement used to improve its 3D consistency.

Conservative mask refinement. A limitation of GEN3C forward warping is background leakage: source-view regions that are occluded, such as the lion body in Fig. 8, may become visible from a novel viewpoint but remain absent from the disocclusion mask. The video diffusion model therefore receives incomplete foreground-completion guidance, as shown in Fig. 8 (second and fourth columns).

We address this failure mode with a conservative masking rule. Because the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unknown, we treat those regions as potentially occupied and construct refined disocclusion masks under that assumption. This refinement allows the video model to account for foreground regions that are not captured by standard point-based rendering.

The procedure constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, following Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ is the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. The resulting surface mesh defines a boundary between visible and invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained by ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ is the original forward-warping disocclusion mask and ϵ is a small tolerance. If the mesh surface is closer than the point-based depth, the corresponding region is conservatively masked as a possible foreground disocclusion. This produces more reliable video-generation guidance under large viewpoint changes, as illustrated in Fig. 8 (third and fifth columns).

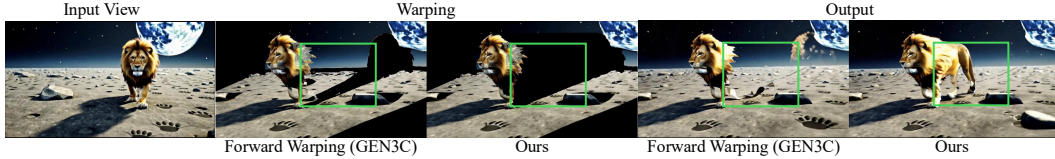


Figure 8: **Rendering function comparison.** The model builds on GEN3C (Ren et al., 2025), which uses forward-warped images for camera-control conditioning. We compare forward warping (left) with our conservative rendering function (right). The highlighted region is incorrectly populated with background pixels under forward warping, whereas our renderer masks it correctly and enables subsequent completion.

B MODEL DETAILS

Progressive training. Directly training the 3DGS decoder on 6 trajectories of 121 frames at 704×1280 resolution is computationally expensive. We therefore use the progressive training schedule in Tab. 3. The final static pre-trained model initializes the dynamic model. Training requires 6 days on 8 NVIDIA A100 80 GB GPUs. We use gsplat (Ye et al., 2025) as the 3DGS implementation.

Subsampling Gaussians. Pixel-aligned 3D Gaussian prediction scales linearly with spatial and temporal resolution. A full representation of 726 frames at 704×1280 resolution would contain 654,213,120 per-pixel Gaussians. We therefore generate one 3D Gaussian per 8×8 spatial neighborhood, reducing the count by a factor of 64 to 10,222,080. Opacity-based pruning further reduces the representation to 2,044,416 Gaussians, yielding a more compact output.

Joint Transformer-Mamba blocks. The reconstruction network uses the block design from Long-LRM (Ziwen et al., 2024): each block contains one Transformer layer followed by seven Mamba-2

Table 3: **Progressive training configuration.**

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1–6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

layers. We repeat this block twice, resulting in 16 layers with hidden dimension 512. This network is substantially smaller than Long-LRM (Ziwen et al., 2024) and Wonderland (Liang et al., 2025a), which use 24 layers with hidden dimension 1024. Although pure Transformer architectures can be expressive, they are substantially slower than Mamba-2 during training and inference. Because both configurations were trained under the same budget, we hypothesize that the Transformer-only variant would require many more iterations to match the rendering quality of the joint design.

Camera encoding. We encode cameras with Plücker embeddings, following prior work (Kant et al., 2024; Zhang et al., 2024e). The embeddings are computed at full spatial and temporal resolution and then projected into the video latent space with the pre-trained RGB encoder. Because the RGB encoder accepts 3-channel inputs whereas Plücker embeddings are 6-dimensional, we encode the 3-dimensional ray directions and the 3-dimensional cross products between ray directions and origins separately. The resulting latent encodings are concatenated along the channel dimension.

Time encoding. The scalar time input is made compatible with the 3-channel RGB encoder by concatenating a 2-dimensional sinusoidal embedding to the original time value along the channel dimension. The values are then replicated over spatial dimensions before being encoded into the video latent space. This construction independently encodes source and target times into the compressed latent representation.

Dynamic data augmentation. Fig. 9 visualizes the dynamic data augmentation procedure for two representative camera trajectories. The procedure constructs paired supervision views that observe the same motion state from different extreme viewpoints.

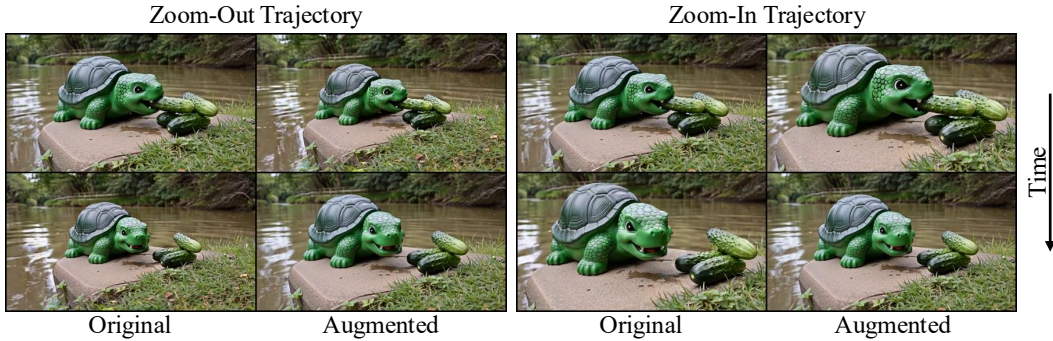


Figure 9: **Dynamic data augmentation videos.** The supervision set is augmented with a motion-reversed video so that each timestep receives broader spatial coverage, preventing low-opacity artifacts at early timesteps. The figure shows two representative trajectories, zoom-out and zoom-in, together with their augmented videos, whose camera motions are temporally inverted.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimer (GEN3C) using five rendered views from generated scenes. The proposed method exhibits fewer artifacts and higher visual fidelity.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare against BTimer (Liang et al., 2025b), a recent method for joint 3D and 4D reconstruction from posed images or videos. We contacted the BTimer authors to run experiments on the same scenes used for *Lyra*. Because BTimer is a regression-based model, we integrate it with the GEN3C (Ren et al., 2025) video diffusion backbone. Specifically, we use the same camera trajectories as our model and provide GEN3C RGB-decoded videos as BTimer inputs. BTimer operates in pixel space and requires 12 input images. As discussed in the main paper, pixel-space models such as BTimer exceed memory limits when directly processing the 726 high-resolution frames generated by GEN3C. We therefore uniformly subsample GEN3C frames. For static evaluation, the 12 selected frames evenly cover the available viewpoints. For dynamic evaluation, 11 frames span the viewpoint range and one additional bullet-time frame corresponds to the reconstructed motion state, ensuring that the target motion is present as a reference. We denote this integrated baseline by BTimer (GEN3C), i.e., BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 reports qualitative comparisons with BTimer (GEN3C), and Tab. 4a reports quantitative results. Our method produces higher visual quality and fewer artifacts.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare with BTimer (GEN3C), defined in Sec. C.1, on 100 out-of-distribution videos from the dynamic *Lyra* dataset. Videos are cropped to 512×512 for a fair comparison because BTimer was primarily trained at that resolution. Tab. 4b shows that our method significantly outperforms BTimer (GEN3C).

C.4 DEPTH VISUALIZATION

Fig. 11 visualizes depth maps predicted from generated 3DGS scenes. Depth supervision reduces flattened geometry without degrading visual quality.

Table 4: Quantitative results for the static and dynamic *Lyra* datasets.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) Quantitative comparison on the static *Lyra* dataset.

(b) Quantitative comparison on the dynamic *Lyra* dataset.

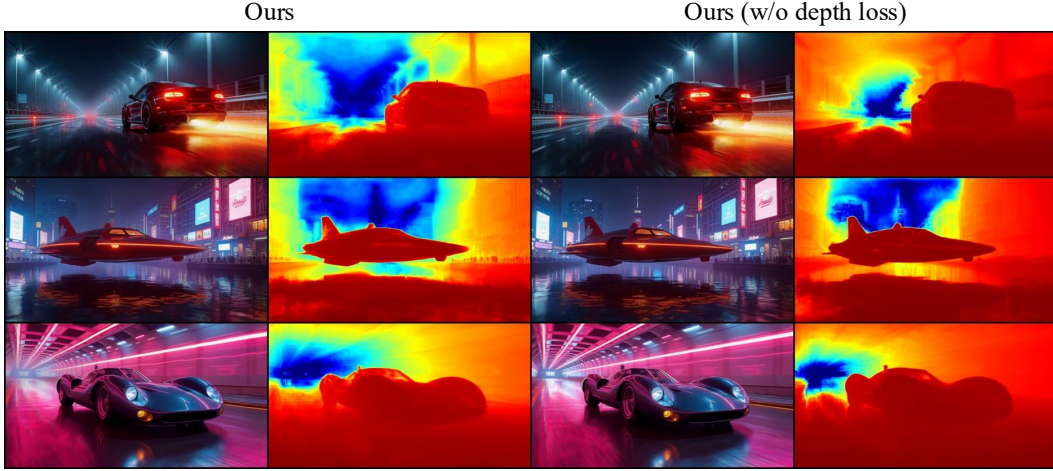


Figure 11: **Depth loss ablation.** The figure reports 3DGS renderings and their associated depth maps. Adding depth supervision reduces flattened geometry while preserving visual quality.

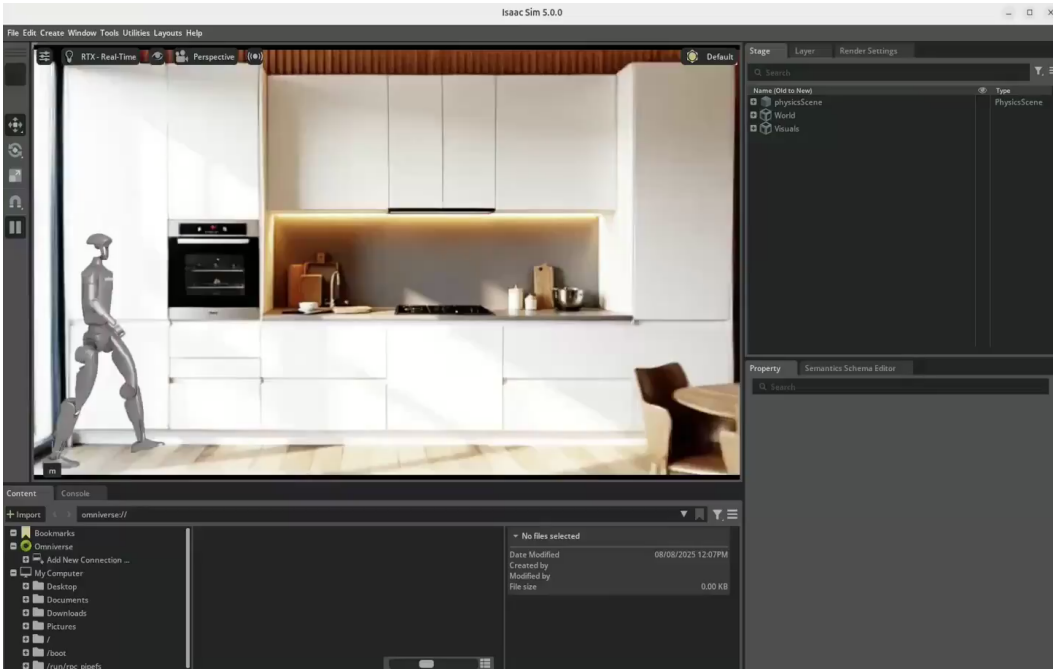


Figure 12: **Robot Simulation.** The figure shows a frame from an Isaac Sim 5.0 robot simulation that uses a 3DGS scene generated by our method as the environment input.

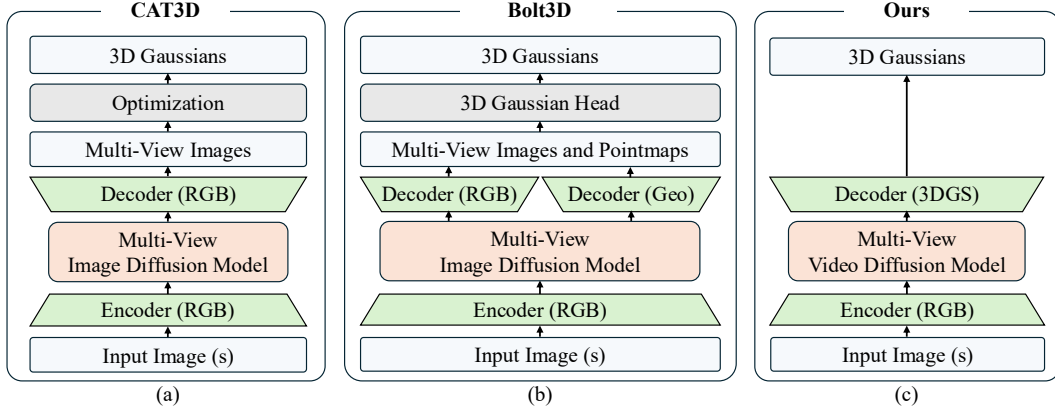


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) generates novel-view images with a multi-view image diffusion model and then reconstructs those images into 3D Gaussians through per-scene optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to emit pointmaps using a geometry autoencoder and predicts the 3D scene with a feed-forward 3D Gaussian head rather than an optimization loop. By contrast, our method uses a pre-trained camera-controlled video diffusion model and *directly* decodes multi-view video latents into 3D Gaussians.

D ROBOT SIMULATION

A primary motivation for a diverse 3D generator is the ability to synthesize simulation environments for autonomous agents. We implement an end-to-end conversion pipeline: 3D scenes are first generated from text with our model, exported as .ply files, and converted to .usdz format. The .usdz files are produced using 3DGUT (Wu et al., 2025a) and its export function. Each resulting .usdz file can be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment. Robots can then be trained and evaluated under diverse generated conditions. Early demonstration results will be released upon acceptance, and Fig. 12 shows a screenshot of the interface.

E RELATED WORK

This section provides additional related work and context. Fig. 13 summarizes the architectural differences between our method and prior systems such as CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b).

3D generation. Early 3D generation methods primarily addressed single object categories and extended GANs to the 3D domain by using neural renderers as inductive biases (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). The introduction of CLIP-based supervision (Radford et al., 2021) enabled more flexible text-driven generation and editing of 3D assets (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). Diffusion-based approaches subsequently improved output quality by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To improve structural coherence, many methods enforce multi-view consistency by generating or optimizing scenes from multiple perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), whereas others use iterative inpainting for scene synthesis (Höllein et al., 2023; Shriram et al., 2024). A related line lifts 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Complementary work studies feed-forward methods that directly predict 3D content from images or text prompts (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). In contrast to our method, these approaches produce static 3D scenes. Other methods specialize in particular scene classes such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) addresses real scenes but does not handle diverse generated scenes or large viewpoint changes.

4D generation. Recent work has made rapid progress on 4D generation, or dynamic 3D scene synthesis. Many methods use text prompts or images as conditioning inputs. After the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent approaches improved both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

Although text conditioning is common, many alternatives lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Other approaches incorporate physics priors into generation pipelines (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), improve motion controllability with template-driven mechanisms (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b), or target compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another research direction extends 3D GAN frameworks to 4D by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). These models are generally constrained by small-scale, single-category datasets, limiting generalization. Most are also object-centric and do not explicitly model background content. Consequently, their visual fidelity remains below that of recent video generation models, which our framework leverages as teachers.

Camera-conditioned video models. Recent video-generation research increasingly incorporates camera control into diffusion models. MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent methods (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) use Plücker coordinates for camera conditioning, while others follow MotionCtrl by using flattened poses (Wang et al., 2025b). Training-free camera-control methods (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) avoid adding trainable parameters. These methods primarily rely on U-Net-based architectures, while other approaches use low-rank mechanisms for camera control (Zhang et al., 2025b).

Several recent methods continue to extend U-Net-based camera-control pipelines (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while others investigate diffusion transformers. Cheong et al. (Cheong et al., 2024) propose a transformer-based framework, although its scene and visual quality remain limited. DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers but relies on predefined, non-continuous camera trajectories. Other work (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) studies pose and geometry estimation with video DiTs. CAT4D (Wu et al., 2025b) constructs a multi-view video diffusion model by fine-tuning a multi-view backbone. Large-scale synthetic data improves camera-controlled video generation, as shown by SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a), which use transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a space-time diffusion model from scratch for novel-view synthesis from a single input image. A related line edits camera motion in videos with underlying dynamics using camera retargeting (Seo et al., 2025; Ma et al., 2025). Another common strategy is to condition camera-controlled video generation on a spatiotemporal cache, such as point clouds or optical flow (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025).